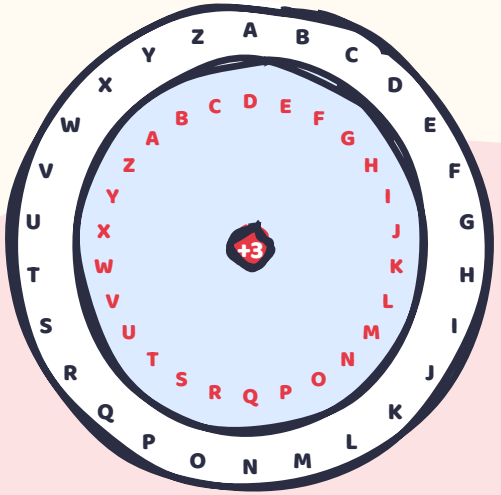
Young Innovators · Class 24

Python Code Breakers: Caesar to Vigenère

Make secret codes. Then break them.

Caesar | Cracking | Vigenère



Did you know?
Julius Caesar shifted letters by 3 to hide army orders over 2000 years ago. Today a computer cracks it in a millisecond.

Let's think...

Can you read this? KHOOR ZRUOG

Intercepted!

We caught a spy's note: **KHOOR ZRUOG**. It looks like nonsense, but every letter follows the same secret rule. By the end of today, your Python program will crack it in a blink.



Random letters?

Another language?

A pattern!

Today: you write Python to **encrypt**, **decrypt** and **crack** codes, from Caesar to the 'unbreakable' Vigenère cipher.

Learning objectives

By the end of this class I can...

1	**Caesar cipher:** encrypt and decrypt with a **Caesar shift** using **ord()**, **chr()** and **% 26**	**Proof:** my program turns HELLO into KHOOR and back
2	**Break it:** crack a Caesar message by **brute force** and by **letter frequency**	**Proof:** I cracked a message without being told the key
3	**Vigenère:** encrypt with a **keyword** (Vigenère) and explain why it beats frequency analysis	**Proof:** my partner could only read my message with the keyword

LEARN

Objective 1 · Caesar cipher · video

Video time!



The Caesar cipher | Journey into cryptography

Khan Academy · about 3 minutes · youtu.be/sMOZf4GN3oc

Watch for...

1. How does the Caesar **shift** work?
2. Why is the Caesar cipher **easy to break**?
3. How many possible **keys** are there?

Scan to watch

After the video: Decode this with a shift of 1: IFMMP. What did it say?

Objective 1 · Caesar cipher

Caesar: shift every letter

The idea

Pick a **key** (shift), e.g. 3: A → D, B → E, … X → A, Y → B, Z → C (wrap around).

In Python, letters are numbers:
ord('A') = 65, **chr(65) = 'A'**
Position: **p = ord(c) − 65** (A=0 … Z=25)
Encrypt: **(p + key) % 26**
Decrypt: **(p − key) % 26**

% (mod) gives the remainder, so 25 + 3 = 28 → 28 % 26 = 2 = C.

Example: a clock



H → K

Z → C

It's 11 o'clock; 3 hours later it's 2, not 14. Clocks wrap around at 12; the alphabet wraps around at 26. That's what % does.

Objective 2 · Break it

Breaking Caesar: two attacks

The idea

Brute force: there are only **25** possible keys. Try them all and look for real words. A computer does it instantly.

Frequency analysis: in English, **E** is the most common letter (~13%), then T, A, O. Count letters in the secret message: the most common one is probably E. If it's H, the shift is 3!

Invented by **al-Kindi** in Baghdad, about 850 CE.

Ethics

Only crack messages made for practice, or with permission. Reading other people's private messages is wrong and often illegal.

Example: guessing a game of hangman



In hangman, everyone guesses E, T, A first because they are the most common. Code breakers use the same trick.

Objective 3 · Vigenère

Vigenère: a different shift each letter

The idea

Use a **keyword**, e.g. **KEY** = shifts 10, 4, 24, repeating.

H+K, E+E, L+Y, L+K, O+E → **RIJVS**

The two L's become **J** and **V**! Same letter, different output, so simple frequency counting fails. Called 'le chiffre indéchiffrable' for 300 years, until Babbage and Kasiski cracked it in the 1800s.

Today we use **AES**: maths so strong all computers on Earth couldn't brute-force it.

Example: a combination lock with many wheels

H	E	L	L	O	text
K	E	Y	K	E	key
R	I	J	V	S	cipher

One wheel has 10 options. Three wheels: 1000. A longer keyword is like adding wheels: the number of possible keys explodes.

Real world

Codes that protect us



HTTPS padlock
Your bank and school portal use AES



WhatsApp
End-to-end encryption on every message



Enigma
Breaking it in WWII shortened the war; Alan Turing helped



Al-Kindi
First known book on frequency analysis, 9th-century Baghdad

? If encryption is so strong today, how do hackers still steal accounts?

Humans are the weak link
Most hacks trick people (phishing, weak passwords), not the maths.

Quick quiz

Cipher maths

Question	Answer
1. Caesar key 3: encrypt CAT.	
2. 27 % 26 = ?	
3. How many Caesar keys must brute force try?	
4. Most common English letter?	
5. Why does Vigenère beat frequency analysis?	

Answers

Cover! 1: FDW · 2: 1 · 3: 25 · 4: E · 5: the same letter encrypts differently

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Cipher wheel race
2. Our code lab
3. Caesar in 10 lines
4. Crack it two ways
5. Vigenère with a keyword
6. Code-breaker's log
7. Cryptographer challenges

MAKE

Class game

Cipher wheel race

How to play

1. Each pair builds a paper cipher wheel (printed sheet + split pin).
2. Team A encrypts a 5-word school motto with a secret Caesar key.
3. Swap messages. Team B must crack it WITHOUT the key: count letters or try shifts.
4. First team to read the motto wins. Then try a Vigenère round with a 3-letter keyword!

Then discuss

Which was faster: trying all keys, or counting letters? Which helped for Vigenère?



Big idea

Short messages are hard to count; long ones give the secret away.

MAKE

Build it

Our code lab

Thonny
Python 3, no extra packages

Cipher wheel sheet
Print, cut, join with a split pin

ASCII card
A=65 … Z=90

Mystery messages
messages.txt from the class folder

A partner
To swap and crack

Timer
For the cracking race

Save as ciphers.py. Every function we write today goes in the same file so you can reuse caesar() later.

Practice only
Only crack messages made for this class. Never try to read someone's private messages or accounts.

MAKE

Objective 1 · Caesar cipher · Build step 1 of 3

Caesar in 10 lines

```
def caesar(text, key):
    out = ""
    for c in text.upper():
        if c.isalpha():
            p = ord(c) - 65              # A=0 ... Z=25
            out += chr((p + key) % 26 + 65)
        else:
            out += c                     # keep spaces, punctuation
    return out

secret = caesar("HELLO WORLD", 3)
print(secret)                  # KHOOR ZRUOG
print(caesar(secret, -3))      # decrypt: shift back
```

Trick

Decrypting is just encrypting with the negative key!

isalpha()

Only letters are shifted, so spaces stay as spaces.

Try

Ask for input(): message and key from the user.

MAKE

Objective 2 · Break it · Build step 2 of 3

Crack it two ways

```
from collections import Counter

msg = "PHHW PH EHKLQG WKH VWUHHW WKUHH WUHHV"

# Attack 1: brute force
for key in range(1, 26):
    print(key, caesar(msg, -key))

# Attack 2: frequency analysis
letters = [c for c in msg if c.isalpha()]
top = Counter(letters).most_common(1)[0][0]
key = (ord(top) - ord("E")) % 26
print("Most common:", top, "→ guess key", key)
print(caesar(msg, -key))
```

Brute force

Scan the 25 lines with your eyes. Which one is English?

Counter

Counter counts every letter. most_common(1) gives the winner.

Limits

Frequency guessing needs a long-ish message. Try a 3-word one: does it fail?

MAKE

Objective 3 · Vigenère · Build step 3 of 3

Vigenère with a keyword

```
def vigenere(text, keyword, direction=1):
    out, i = "", 0
    keyword = keyword.upper()
    for c in text.upper():
        if c.isalpha():
            shift = ord(keyword[i % len(keyword)]) - 65
            out += caesar(c, direction * shift)
            i += 1                          # next key letter
        else:
            out += c
    return out

s = vigenere("HELLO", "KEY")
print(s)                            # RIJVS
print(vigenere(s, "KEY", -1))       # HELLO
```

Reuse

Vigenère is just Caesar with a changing key. We call our own caesar()!

i % len

Makes the keyword repeat: K, E, Y, K, E, Y...

Test

Run Attack 2 on a Vigenère message. Does the frequency trick still work?

MAKE

Test like an engineer

Code-breaker's log

Test	Result	Notes
caesar('HELLO', 3) = KHOOR?		
Decrypt gives the original back?		
Cracked msg: key found		
Vigenère 'HELLO','KEY' = RIJVS?		
Frequency attack on Vigenère?		

Debug checklist

Lowercase letters messed up? We use .upper() first.
Wrong letters near Z? Check you used % 26.
IndexError in Vigenère? Use i % len(keyword).

A good test always includes edge cases: Z, spaces, empty message, key 0, key 26.

MAKE

Level up!

Cryptographer challenges

Choose your challenge

Auto-crack: score each brute-force line by counting common words (THE, AND) and print only the best.

Kasiski: research how repeated chunks reveal a Vigenère keyword length.

Atbash: write the Hebrew Atbash cipher: A ↔ Z, B ↔ Y...

Urdu? how would a Caesar cipher work for the Urdu alphabet?

Think like an engineer

Why is a long, random keyword much stronger than 'CAT'?

Boss level

Write a one-time pad: a random key as long as the message. Why is it truly unbreakable?

Objective check

Can I really do it? Prove it.

01	Encrypt your name with Caesar key 5, by hand and with your program.	○ ○ ○
02	Crack the teacher's mystery message. Which attack did you use?	○ ○ ○
03	Explain why Vigenère beats frequency analysis, using the two L's in HELLO.	○ ○ ○

Rate yourself

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

Teacher: Cipher challenge: the teacher posts 3 messages (Caesar, Caesar short, Vigenère). Which groups crack which?

Show what you know

What did we learn?

- [x] ord() and chr() turn letters into numbers and back
- [x] % 26 wraps the alphabet around
- [x] Caesar falls to brute force (25 keys) or E-counting
- [x] Vigenère uses a keyword: a different shift per letter
- [x] Real security uses AES, and strong human habits

At home
Send a family member a Caesar message and see if they can crack it.



Exit ticket: what is the Caesar key if E became J?

Next class:
Robot Sumo Challenge!